COM 220: SOFTWARE ENGINEERING

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Self Introduction

Name?
Why did you choose this course?

What are your motivations?
Role model in Software Engineering field?

Do you have ideas on a particular social problem?
Do you think the course will enable you implement your idea?





About Profession

A Physician, a Civil Engineer and a Computer Scientist were arguing about what was the oldest profession in the world.

- The Physician remarked,
  "Well, in the Bible, it says that God created Eve from a rib taken out of Adam. This clearly requires surgery, and so I can rightly claim that mine is the oldest profession in the world."
- The Civil Engineer interrupted, and said,
  " But even earlier in the book of Genesis, it states that God created the order of the heavens and the earth from out of the chaos. This was the first and certainly the most spectacular application of civil engineering. Therefore, fair doctor, you are wrong; mine is the oldest profession in the world.“
- The Computer Scientist leaned back in the chair, smiled and then said confidently,
  "Ah, but what do you think created the chaos ? "



Course Objectives

By the end of the course, the learner should be able to:

(a)Understand the main ideas and issues of software engineering, including the role of software development processes.

(b)Apply several current software engineering techniques, including UML.

(c)Gain experience of developing software in teams and acting professionally.

(d)Contribute to industrial software development projects.



Software Engineering

Software Engineering

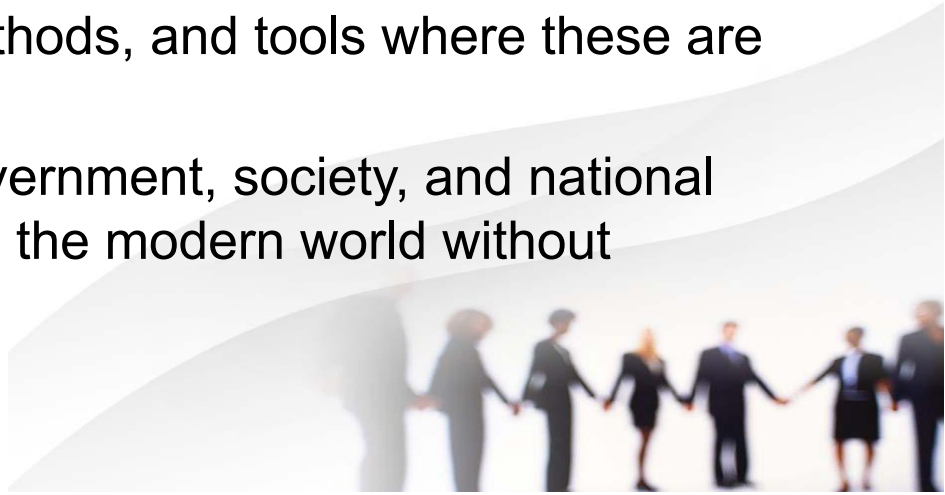
– Software engineering is an engineering discipline that is concerned with all aspects of software production from the early stages of system specification through to maintaining the system after it has gone into use.

Key phrases:

– Engineering discipline

Engineers make things work. They apply theories, methods, and tools where these are appropriate.

Software engineering is essential for the functioning of government, society, and national and international businesses and institutions. We can't run the modern world without software.



Software Engineering

Software Engineering is a collection of techniques, methodologies and tools that help with the production of:

- a high quality software system
- with a given budget
- before a given deadline

while change occurs.

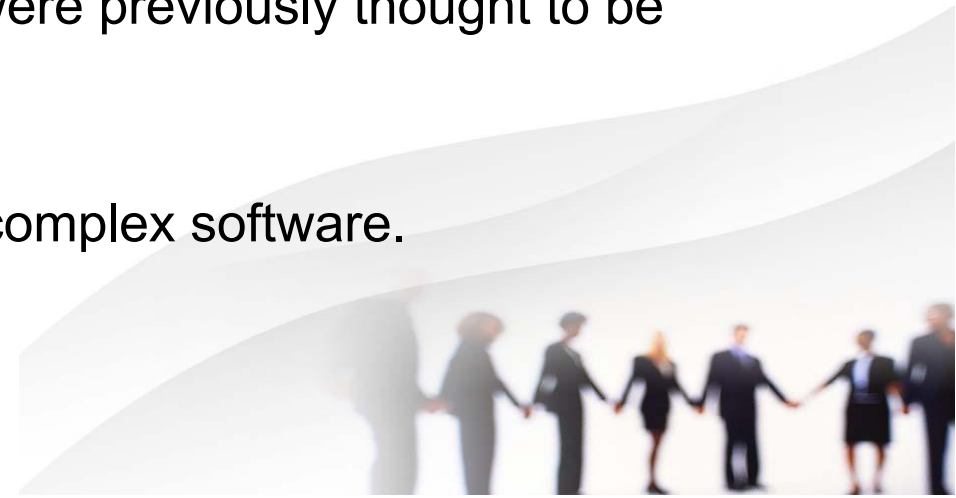


Why do Softwares Fail?

Software engineering is criticized as inadequate for modern software development due to the following factors:

1. Increasing system complexity

- As new software engineering techniques help us to build larger, more complex systems, the demands change.
- Systems have to be built and delivered more quickly; larger, even more complex systems are
- required; and systems have to have new capabilities that were previously thought to be impossible.
- New software engineering techniques have to be
- developed to meet new the challenges of delivering more complex software.



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2. Failure to use software engineering methods

- It is fairly easy to write computer programs without using software engineering methods and techniques.
- Many companies have drifted into software development as their products and services
- have evolved.
- They do not use software engineering methods in their everyday work. Consequently, their software is often more expensive and less reliable than it should be.
- We need better software engineering education and training to address this problem.



Software Engineering ethics

Confidentiality

You should normally respect the confidentiality of your employers or clients regardless of whether or not a formal confidentiality agreement has been signed.

Competence

You should not misrepresent your level of competence. You should not knowingly accept work that is outside your competence.

Intellectual property rights

You should be aware of local laws governing the use of intellectual property such as patents and copyright. You should be careful to ensure that the intellectual property of employers and clients is protected.



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Computer misuse

You should not use your technical skills to misuse other people's computers. Computer misuse ranges from relatively trivial (game playing on an employer's machine) to extremely serious (dissemination of viruses or other malware).



SOFTWARE PROCESSES

Lecturer: Wanini KarU



Objectives

understand the concepts of software processes and software process models;

■ have been introduced to three general software process models and when they might be used;

■ know about the fundamental process activities of software requirements engineering, software development, testing, and evolution;

■ understand why processes should be organized to cope with changes in the software requirements and design;

■ understand the notion of software process improvement and the factors that affect software process quality.



Software Processes

Reference Chapter 3 Slides on Process Models

Lecturer: Wanini KarU

